

Answer All the Questions

QUESTION 1: SUPERVISED LEARNING

Import the Iris dataset from sklearn library and keep 75% for the training set

1. Use logistic regression, SVM, Random forest, and decision Tree and get the score for each Model (10 marks)
2. Use the cross_val_score object with 5 slots, and Measure the different scores for logistic regression, SVM, Random forest, and decision Tree. (10 marks)
3. Plot the confusion Matrix for the model with the best accuracy (10 marks)

QUESTION 2: SUPERVISED LEARNING

Import the Digit dataset from sklearn library

1. Using the K-Means, vary the number of Cluster from 1 to 20, plot the K against the SSE (Sum of Squared Errors), and deduce the optimal value of K (10 marks)

Question 3: DEEP LEARNING

Import the *Cifar10* dataset from tensorflow.keras library. Build a Convolutional Neural Network, made up of 03 stacks (Convolution, Relu, Pooling), coupled to a fully connected output layer,

1. Compile, and train it (10 marks)
2. Generate the summary of the network (10 marks)
3. Plot the confusion Matrix for the model (10 marks)